

Designed by Remus Teodorescu 2026

LECTURE 10.2

Electrolysers

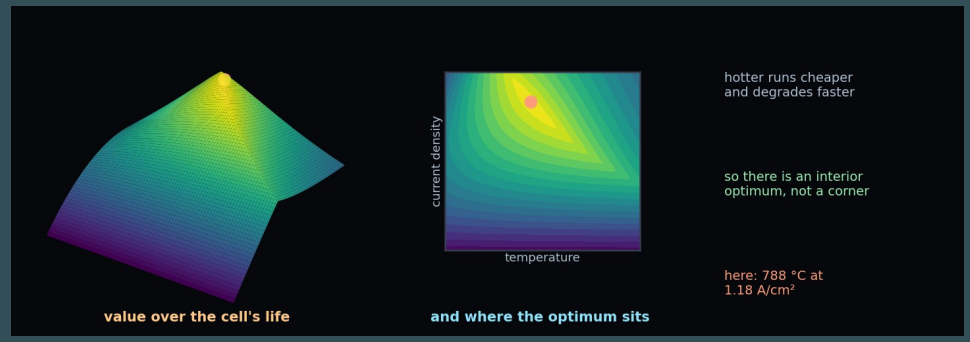
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

STEAM · VOLTAGE · LIFETIME



Schematic: the value a cell produces over its lifetime against temperature and current, with its optimal point in the interior - hotter is cheaper per hour and ages faster. The lecture sets out the electrolyser and the thermoneutral voltage, solves the fuel channel, and finds the optimal voltage by a gradient through one PINN trained over the whole operating range.

Learning Objectives

SECTIONS, IN ORDER

1. **The cell** — three electrolysers, the losses, the thermoneutral voltage

2. **The channel** — steam, current and heat settling together

3. **The optimal voltage** — one PINN for every operating point, the cost and its gradient

Incropera carries the transport, Liu the optimisation through a network. The example is Ex_10.2's electrolysis cell: the channel of L9.2, now with its electrode. This is the last physics lecture, and the place Part 2 has been heading.

READ, AND WHAT EACH ANSWERS

Incropera ch. 1	where do the losses sit?
Incropera ch. 14	how does the steam run short?
Liu ch. 5	what does a gradient buy?

Three Electrolysers

Water electrolyser — a cell that splits water into hydrogen and oxygen with electricity. The three industrial types differ in their electrolyte, and so in the ion that carries the charge and in the temperature at which they run.

ONE ELECTROLYTE EACH

alkaline — a potassium hydroxide solution near 80 degrees; OH^- carries the charge and a diaphragm keeps the gases apart

PEM — a humid polymer membrane near 70 degrees; H^+ crosses to the hydrogen side and liquid water is split at the oxygen side

solid oxide — a ceramic near 800 degrees; O^{2-} crosses to the air side, steam is split and heat supplies part of the energy

COST AND CONTROLLABILITY

	alkaline	PEM	solid oxide
temperature [°C]	70 – 90	50 – 80	700 – 850
energy [kWh/kg H_2]	50 – 78	50 – 83	45 – 55
cost [USD/kW]	500 – 1000	700 – 1400	stack >2000
minimum load [%]	10 – 30	10 – 40	about 50
ramp [% per s]	0.5 – 5	3 – 10	about 0.2
start from cold	minutes	minutes	hours: kept hot

Typical values, rounded; costs are IRENA's 2020 system prices (installed plants more: 2000 – 2600 USD/kW outside China, 600 – 1200 in China; IEA 2025). The solid oxide rows are one product's data sheet, and its limits are thermal, not electrical: the stack follows the current within seconds, and designs that manage the heat, such as Dynelectro's AC:DC cycling, reach low minimum load.

IRENA (2020); Dorn et al. (2026); IEA (2025)

Electrolysis at Industrial Scale

Power-to-X — the conversion of renewable electricity into hydrogen by electrolysis, and of the hydrogen into a fuel or chemical, such as methanol, that can be stored and shipped.

KASSØ, DENMARK, SINCE 2025

electrolysers	52 MW of PEM, from Siemens Energy
power	a 304 MW solar park beside the plant
product	up to 42 000 t of e-methanol a year

INSTALLED WORLDWIDE, AND 2030

installed	0.6 GW in 2021, 2 GW in 2024, 4.2 GW in 2025
by technology	in 2022: alkaline 60 %, PEM 30 %, SOEC under 1 %
by 2030	26 GW committed; 520 GW announced, much now out of reach
China	65 % of the capacity installed or committed



Laura Mærsk, e-methanol fuel
Jens Dohrn, CC BY-SA 4.0



2 MW PEM module, ITM Power
Bubble60, CC BY-SA 4.0

WHY ALKALINE STILL LEADS

mature	a century in industry
cheap	no noble metals, large stacks
China's makers	alkaline at 600 – 1200 USD/kW

Elsewhere 2000 – 2600 USD/kW (IEA). More: Kassø, ITM Power.

The Solid Oxide Electrolyser

Solid oxide electrolysis cell — a ceramic electrochemical cell that splits steam into hydrogen and oxygen near 800 degrees. The name comes from its electrolyte, a solid oxide (yttria-stabilised zirconia) that conducts only O^{2-} ions. In reverse it is a solid oxide fuel cell; the electrodes are therefore the fuel and air electrode, as anode and cathode swap between modes.

WHY THE HIGH TEMPERATURE PAYS

$$\Delta H = \Delta G + T\Delta S, \qquad \Delta G \downarrow \text{ as } T \uparrow$$

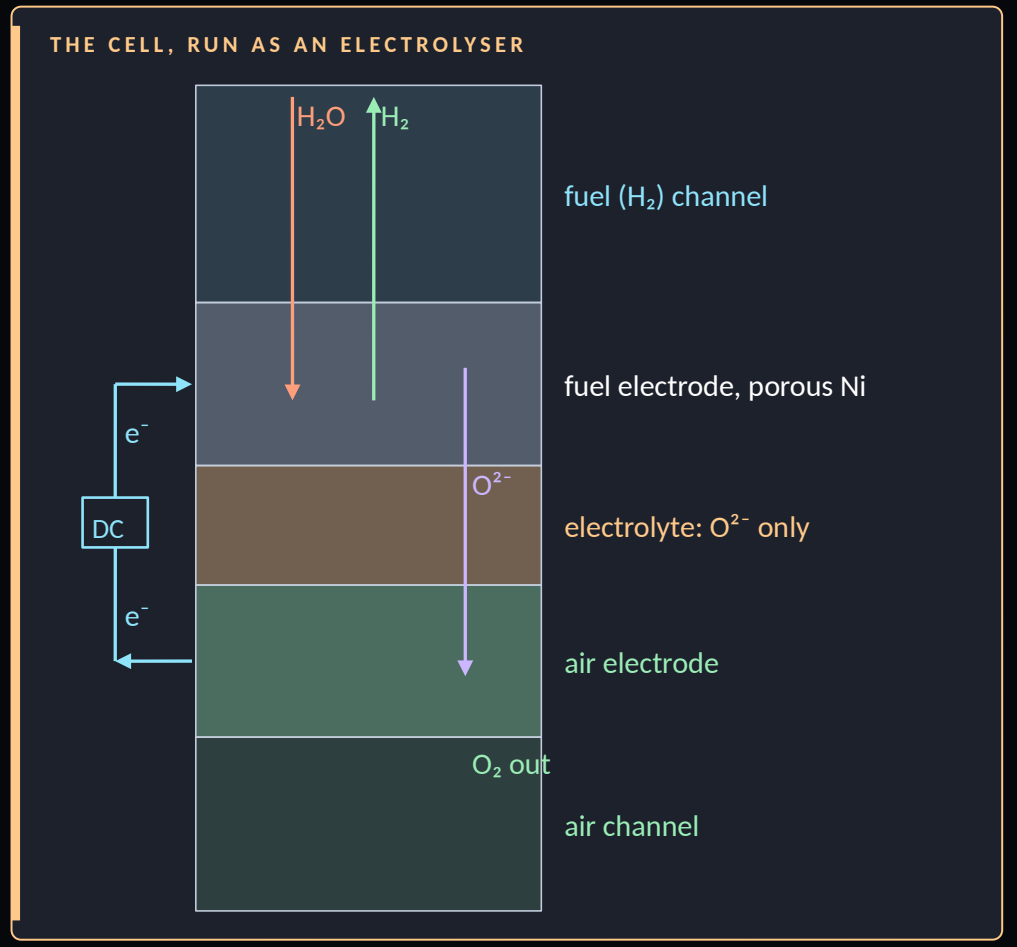
ΔH the energy to split water, ΔG the part that must be electricity, $T\Delta S$ the part heat can supply

THE THREE LAYERS, AND THEIR REACTIONS

fuel electrode — $H_2O + 2e^- \rightarrow H_2 + O^{2-}$: two electrons from the supply; the oxide ion enters the ceramic

electrolyte — O^{2-} hops through oxygen vacancies in the hot ceramic; electrons and gases cannot pass

air electrode — $2\,O^{2-} \rightarrow O_2 + 4e^-$: the ions return their electrons to the circuit and leave as oxygen



The Polarisation Curve

Nernst potential — the voltage at which no current flows: the reversible voltage of the reaction, set by the local gas composition and the temperature; it rises as the steam is used up. The polarisation curve is this potential plus the losses, which grow with the current.

THE NERNST POTENTIAL, AND THE LOSSES ON TOP

$$E = E^0(T) + \frac{RT}{2F}\ln\left(\frac{(1-y)\,p_{O_2}^{1/2}}{y}\right)$$

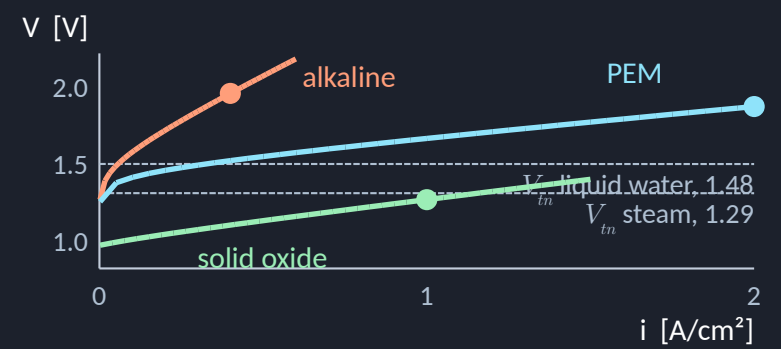
$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

E the Nernst potential, y the steam fraction, p_{O_2} the air side's oxygen, V the cell voltage
R the gas constant, T the temperature, F Faraday's constant, 2 the electrons per molecule
$\eta_{act}, \eta_{ohm}, \eta_{conc}$ the activation, ohmic and gas-transport losses; in electrolysis they add

Ex_10.2 keeps the losses separate: activation, a resistance that falls with temperature, and gas transport with its limiting current - every constant an estimate, flagged in the notebook.

THREE CELLS: VOLTAGE TO ENERGY PER KG

SCHEMATIC, TYPICAL VALUES



alkaline	1.94 V at 0.4 A/cm²: 52 kWh/kg
PEM	1.85 V at 2.0 A/cm²: 49 kWh/kg
solid oxide	1.25 V at 1.0 A/cm²: 33 kWh/kg

One volt costs 26.6 kWh/kg (stack); a system needs 45 – 55.

The Thermoneutral Voltage

Thermoneutral voltage — the operating voltage at which the heat the losses release equals the heat the reaction absorbs, so the net heat is zero. For steam near 800 degrees it is 1.29 volts; stacks are operated at or near it, and with outside heat a stack can run below it, endothermically.

THE BALANCE, AND THE HEAT IN ONE TERM

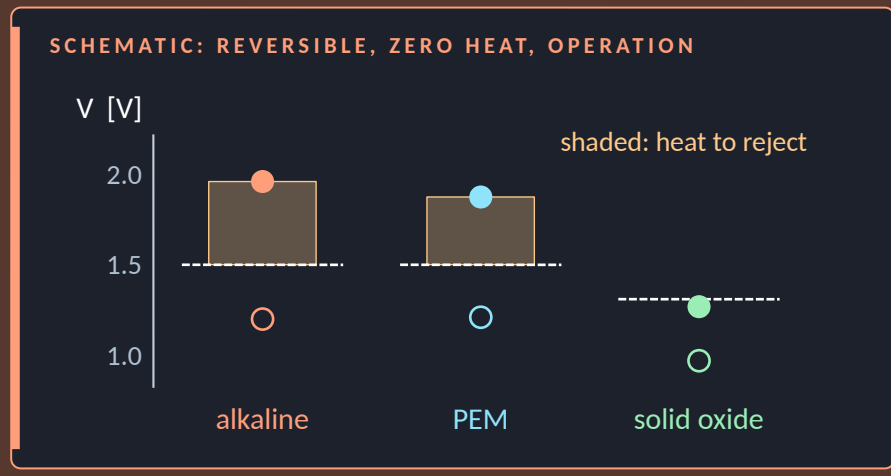
$$V_{tn} = \frac{\Delta H}{2F} = 1.29\ \text{V} \quad (\text{steam},\ 800\ ^\circ\text{C})$$

$$q = i(V - V_{tn})$$

V_{tn} the thermoneutral voltage, ΔH the enthalpy of splitting a mole of steam
$2F$ the charge of its two electrons (F Faraday's constant), q the heat per electrode area, i the current density

TALK · 2 MIN How would you explain, in your own words, where the heat of an electrolyser comes from and where it goes? Then, in the drawing on the right, why do the two cold cells reject heat while the solid oxide cell runs at its dashed line?

THREE CELLS ON ONE VOLTAGE AXIS



alkaline, PEM 0.4 – 0.5 V above: heat to remove

solid oxide at V_{tn}: no net heat in the stack

Liquid water: 1.48 V includes evaporation. Steam at 800 °C: 1.29 V, the steam raised outside.

The Fuel Channel

Fuel channel — the hydrogen channel of L9.2, run as an electrolyser: 90 % steam enters at 1 m/s and the electrode takes steam and returns hydrogen, so the steam mole fraction y, the share of steam molecules in the gas, falls along the channel.

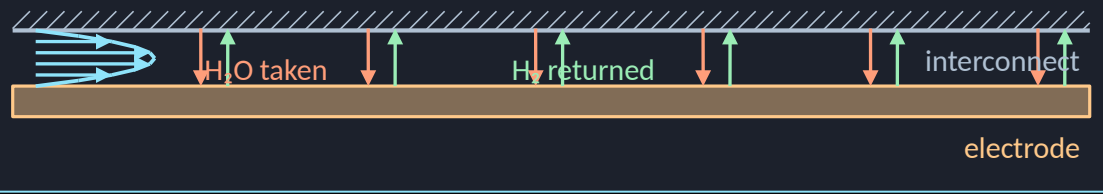
THE TWO SCALED BALANCES OF EX_10.2

$$\frac{dy}{dx} = \frac{1}{Pe}\frac{d^2y}{dx^2} - K_I\, i, \qquad Pe = 125$$

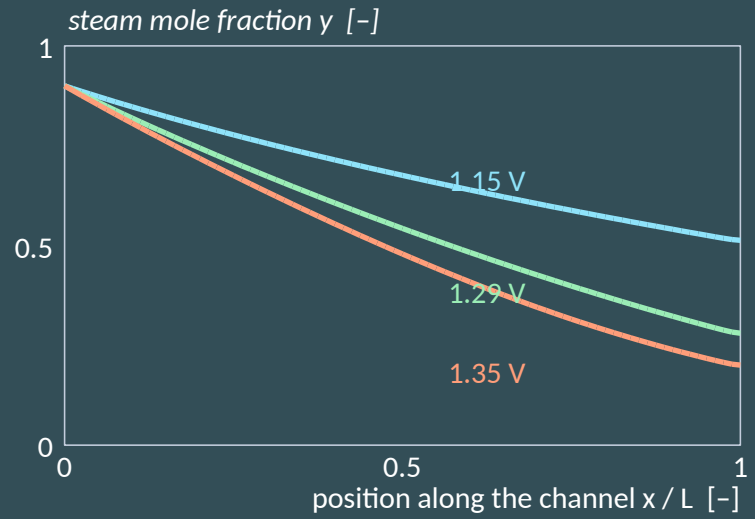
$$\frac{d\theta}{dx} = \frac{1}{Pe_T}\frac{d^2\theta}{dx^2} + S_Q\, i(V - V_{tn}), \qquad Pe_T = 16$$

y the steam mole fraction, $\theta = (T - T_{in})/50$ K the scaled temperature, i the local current density [A/cm^2]
Pe, Pe_T the Péclet numbers; K_I, S_Q gather u, h, c, F, Γ, L; $V_{tn} = 1.29$ V
$i > 0$ in electrolysis, so $-K_I\, i$ makes y fall; hydrogen is $1 - y$, exchanged one for one : no equation of its own

STEAM TAKEN, HYDROGEN RETURNED



STEAM ALONG THE CHANNEL



THE CHANNEL IN THE THREE CELLS

	fed	the channel must
alkaline	KOH solution, in excess	carry gas bubbles out
PEM	liquid water, in excess	carry bubbles, cool
solid oxide	steam, 69 % used	keep steam to the outlet

The Local Current

Local current density — the current per electrode area at one point. The cell is held at one voltage V; the local steam fraction and temperature set the Nernst potential, the kinetics and the resistance, and the current is whatever satisfies the charge transfer there.

ONE VOLTAGE, A CURRENT THAT VARIES

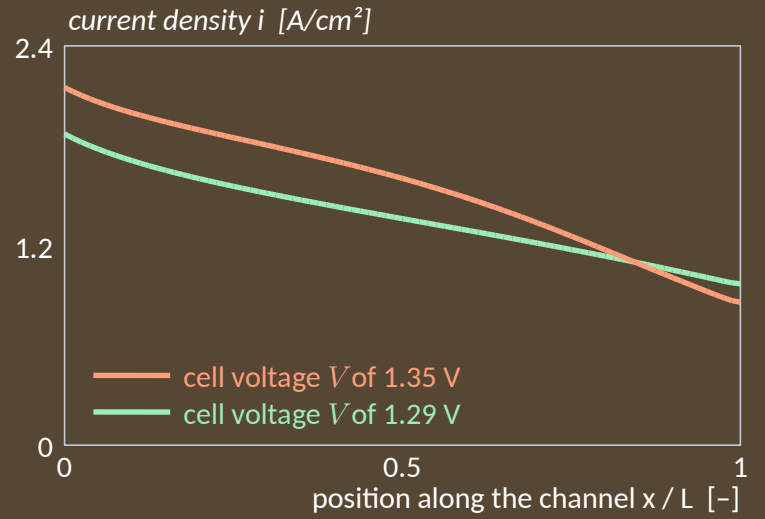
$$V = E(y,T) + \frac{RT}{F}\operatorname{asinh}\frac{i}{2i_0} + i\,ASR(T) - \frac{RT}{2F}\ln\left(1 - \frac{i}{i_L(y)}\right)$$

$E(y,T)$ the Nernst potential of the local gas, i_0 the exchange current density, $ASR(T)$ Arrhenius (estimates)
$i_L(y)$ the limiting current, proportional to y; no derivative: an equation for i at every point
the three terms are the polarisation curve's $\eta_{act}, \eta_{ohm}, \eta_{conc}$, written out; $i < i_L$ keeps the last positive

WHY IT FALLS, AND WHAT IT MEANS

1 · steam used	the steam fraction y falls along the channel
2 · E rises	the Nernst potential grows as y falls
3 · i falls	and so does the heat it makes or takes
4 · T answers	the resistance follows the temperature
→ multiphysics	no field can be solved first and handed on

THE CURRENT ALONG THE CHANNEL



THE CURRENT IN THE THREE CELLS

	along the channel	set by
alkaline	nearly even	gas bubbles
PEM	nearly even	bubbles, membrane
solid oxide	1.87 to 0.97 A/cm²	steam used up

Finite Volumes Along the Channel

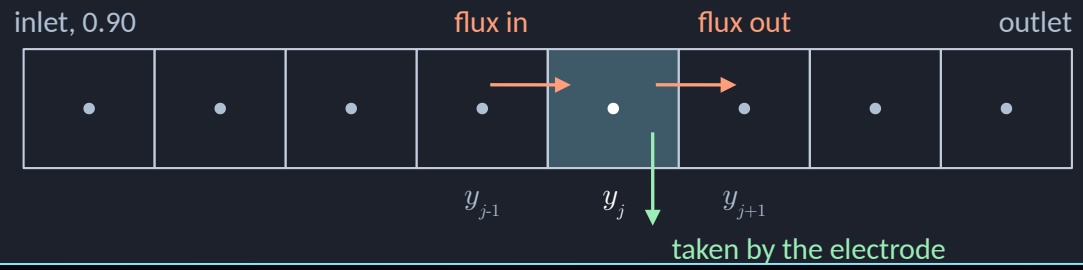
Finite volumes (FVM) — the channel cut into cells, and for each cell the balance of its steam and of its heat: what flow and diffusion carry in through one face, less what leaves by the other, equals what the electrode takes or the current releases. Shared faces conserve both exactly.

THE BALANCE OF CELL j

$$\frac{y_j - y_{j-1}}{h} - \frac{y_{j+1} - 2y_j + y_{j-1}}{Pe\,h^2} + K_I\, i_j = 0, \qquad \text{and its twin for } \theta_j, \qquad j = 1, \ldots, n$$

y_j, θ_j steam and scaled temperature in cell j, $h = 1/n$, i_j the root of the charge transfer in cell j
$2n$ equations, solved together by Newton-Raphson; the coupling blocks carry $\partial i/\partial y$ and $\partial i/\partial \theta$

EIGHT OF THE n CELLS, CELL j PICKED OUT



REFINED, AGAINST THE REFERENCE

cells	steam	kelvin	
25	0.0131	0.55	
50	0.0052	0.29	meets 0.01, 0.5 K
100	0.0024	0.14	
200	0.0015	0.07	

The error falls about in proportion to the cell width, because the flow term takes the upstream cell. One solve is a handful of Newton-Raphson steps on the 2n unknowns with their two coupling blocks, some 25 ms, repeated for every operating point.

One PINN for Every Operating Point

Parametric PINN — a PINN trained over three domains at once: the position x along the channel, the cell voltage V from 1.15 to 1.35 V and the inlet temperature from 750 to 850 degrees. After this one training it returns all three fields at any operating point in that range.

THE PHYSICS IN THE TRAINING

steam balance

$$\frac{\partial y}{\partial x} = \frac{1}{Pe}\frac{\partial^2 y}{\partial x^2} - K_I\, i$$

energy balance

$$\frac{\partial \theta}{\partial x} = \frac{1}{Pe_T}\frac{\partial^2 \theta}{\partial x^2} + S_Q\, i(V - V_{tn})$$

charge transfer

$$V_{cell}(i, y, T) = V, \qquad i = i_L(y)\,\sigma(\mathcal{N}_3)$$

boundaries

$$y = 0.90 + x\,\mathcal{N}_1, \quad \theta = x\,\mathcal{N}_2; \qquad \frac{\partial y}{\partial x} = \frac{\partial \theta}{\partial x} = 0 \text{ at } x = 1$$

$$\mathcal{L}(\theta) = \overline{r_y^2} + \overline{r_\theta^2} + \overline{\left(\frac{V_{cell} - V}{50\ \text{mV}}\right)^2} + \lambda\,\overline{\text{outlet slopes}^2}$$

r_y, r_θ the residuals of the two balances at points (x, V, T_{in}) over 1.15 to 1.35 V and 750 to 850 °C
the charge residual in units of 50 mV, so millivolts weigh like the balances; λ the outlet-slope weight

WHAT THE ONE TRAINING BUYS

one point	FVM in 25 ms is the tool
1000 points	one batch, about 500 times faster
then	the cost $C(V)$ and dC/dV
accuracy	0.0022, 0.19 K; FVM 0.0068, 0.40 K

THE TRIAL FUNCTION IN PYTORCH

```
N1,N2,N3 = MLP(3,3,64,4)(p)   # x, V, T_in in
y = Y_IN + x*N1; th = x*N2    # inlets exact
i = i_lim(y)*sigmoid(N3)      # 0 <= i < i_L
```

Trained once, offline: 33 min on a CPU, repaid after some 80 000 operating points.

Optimisation of the Cell Voltage

Cost function — the number the operator minimises: electricity bought, less hydrogen sold, plus the share of the stack aged in the hour, as a function of the cell voltage V. The prices and the ageing law are assumptions; the optimum is conditional on them.

THE COST, ITS UNITS, AND ITS MINIMUM

$$C(V) = p_{el}\,\frac{V\,I}{1000} - p_{H_2}\,\frac{3600\,I\,M_{H_2}}{2F} + \frac{c_{stack}\,P_{rated}}{\Delta ASR_{EOL}}\,\Delta ASR_h$$

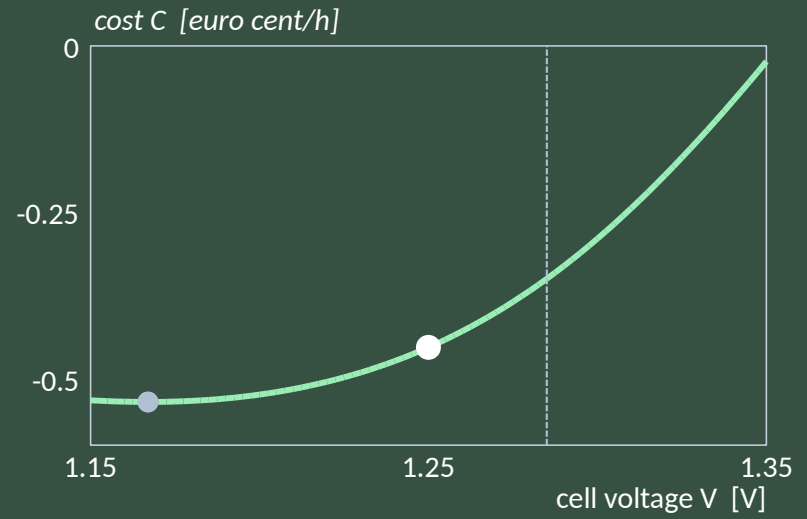
$$I = A\int_0^1 i\,dx \;\;[\mathrm{A}], \qquad \frac{V\,I}{1000}\;[\mathrm{kW}], \qquad \frac{3600\,I\,M_{H_2}}{2F}\;[\mathrm{kg/h}], \qquad V^* = \arg\min_V C \;\; \text{s. t.} \;\; \max_x |T - T_{in}| \le 10\ \mathrm{K}$$

C [cent/h] (negative : a profit); $p_{el} = 0.05$ euro/kWh, $p_{H_2} = 5$ euro/kg : assumptions; the 1000 converts W to kW the 3600 converts per second to per hour; ΔASR_h the resistance grown in the hour (the assumed law below)

THE OPTIMAL VOLTAGE IN PYTORCH

```
V = torch.tensor(1.25, requires_grad=True)   # the decision variable
J = C(V) + mu * relu(dT(V) - 10.0)**2        # the cost, the limit as a penalty
J.backward(); V -= lr * V.grad               # downhill through the network
```

THE COST AGAINST THE VOLTAGE



free optimum	1.167 V, but the cell cools 25 K
with the limit	V^* = 1.250 V, 0.45 cent/h earned
checked by	a sweep of 81 solver runs

No Optimal Point Over a Lifetime

Degradation — the slow rise of the cell's area-specific resistance over thousands of hours, from changes in its layers (right). The rate law below is an empirical form; its constants in Ex_10.2 are assumed, and the grown resistance only prices the hour - it is not fed back into the model.

THE DEGRADATION LAW, AND THE LIFETIME OBJECTIVE

$$\frac{dASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right) |i|^n$$

$$\min_{V(t),\,T(t)} \int_0^{t_{life}} [p_{el} P - p_{H_2} \dot{m}_{H_2}]\, dt \quad \text{s.t.} \quad ASR(t_{life}) \le ASR_{\max}, \quad |\nabla T| \le |\nabla T|_{\max}$$

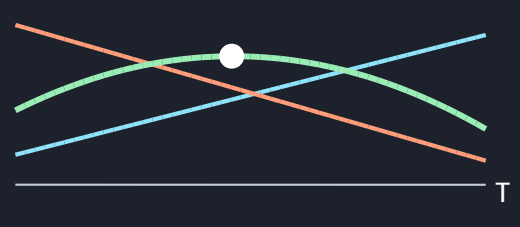
k_0, E_d, n ASSUMED in Ex_10.2, of the right order (published rates span an order of magnitude) an empirical form, Arrhenius in T and a power of i, not derived from the mechanisms on the right

TALK · 2 MIN How would you explain, in your own words, why a device like this has no best operating point? Then, in the drawing on the right, what moves the white dot when the price of electricity falls?

WHAT AGES, AND WHY

layer	mechanism
fuel electrode	nickel grains coarsen and migrate: fewer reaction sites
air electrode	oxygen pressure at the interface lifts it off
steel, air side	chromium vapour deposits and blocks sites

THE TRADE, SCHEMATIC



cent/h: hotter is cheaper

hours left: fewer

product: an optimum inside

so the answer a trajectory $V(t), T(t)$

Operating for Cost and Lifetime

Operating strategy — the cell voltage $V(t)$ and temperature $T(t)$ chosen over the life of the stack to minimise the cost of each kilogram of hydrogen: the electricity bought hour by hour, and the stack replaced when its resistance reaches the end-of-life limit.

THE LEVELISED COST OF HYDROGEN

$$\mathrm{LCOH} = \frac{C_{stack} + \sum_t p_{el}(t)\, P(t)\, \Delta t}{\sum_t \dot{m}_{H_2}(t)\, \Delta t} \quad \text{over the stack's life}$$

LCOH the cost of a kilogram of hydrogen, C_{stack} the stack, replaced at end of life, $p_{el}(t)$ the hourly price $P(t)$, $\dot{m}_{H_2}(t)$ power and hydrogen at the chosen $V(t), T(t)$; the life ends when ASR reaches $ASR_{\max}$

THE TRADE-OFFS

higher voltage	more hydrogen per hour, faster ageing
higher temperature	lower resistance, faster ageing
cheap hours	run harder when the price is low
near V_{tn}	small heat, small gradients in the ceramic

WHAT THE PINN ADDS

one hour	the cost in a millisecond
gradient	dC/dV and dC/dT by autograd
a solver	one solve per variable per step
a lifetime	an optimiser over $V(t), T(t)$
Ex_10.2	seconds, against a sweep of 81 solves

```
C = lcoh(net, V_t, T_t)           # cost over a life
V_t = V_t - lr * grad(C, V_t)[0]  # schedule, downhill
```

Degradation rates are uncertain tenfold: a predicted lifetime ranks strategies, it is not a promise.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
across the channel	the gas profile, the porous electrode	Incropera ch. 14	the 1-D model: +5 – 9 % current; MP10.2A
thermal transients	inertia between operating points	Incropera ch. 5	cycling, start-up, quasi-steady here
degradation records	k_0, E_d, n fitted, not assumed	Liu ch. 5	lifetime; MP10.2B
button-cell fit	kinetics from measured curves	Incropera ch. 1	validation
daily schedule	the hourly price as driver	Liu ch. 5	grid operation; MP10.2B, L12

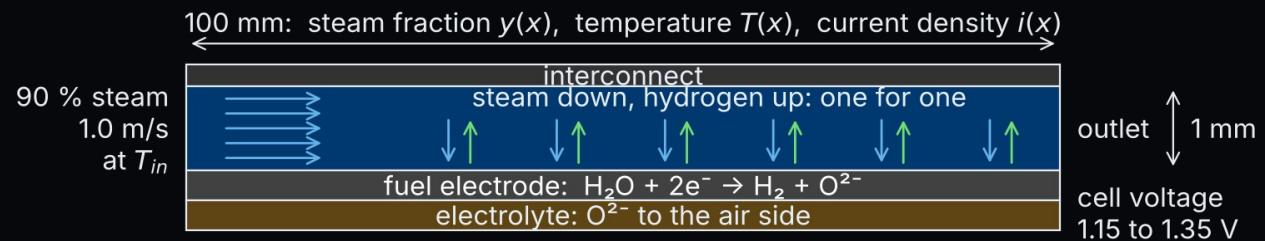
WHY THEY ARE LEFT OUT

Ex_10.2 solves the coupled channel in one dimension, each operating point at its steady state, scored against a boundary-value solver, so none of them is needed. The first and third rows are the two mini projects; the last meets the grid of L12.

Exercise 10.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the electrolyser, its data, the three coupled laws
- **4** — the reference; finite volumes coupled by Newton-Raphson
- **5** — Part A: one PINN for every operating point; 1000 points timed
- **6** — Part B: the optimal voltage under the 10 K limit, by gradient
- **7 – 9** — the report; mini projects MP10.2A and MP10.2B



THE QUESTION EACH PART ANSWERS

1 – 3	which laws couple, and how?
4	what is the coupling worth?
5	one network, every point?
6	which voltage, within 10 K?

REPORT QUESTION, TO CONCLUDE

For which task in this notebook would you use finite volumes, for which the network - and what changes when a voltage must be chosen for every hour of a day? Say which of your numbers rest on estimates.

SUMMARY

Key Takeaways

SOLID OXIDE CELL	
what it is	a ceramic cell, two directions
function	steam to hydrogen, heat helping
train, tune	operated at or near V_{tn}
vs low temperature	a quarter less electricity
used for	storing surplus power as fuel

LIFETIME OPTIMISATION	
what it is	a PINN over voltage and inlet heat
function	three fields, the cost, its gradient
train, tune	once: 1.15 – 1.35 V, 750 – 850 °C
vs a solver	autograd gradient; a solver re-solves
used for	many points, and the choice of V

Next — L11.1: a network in a real-time loop, where there is no PDE at all.

Questions

1 – 2 · A NETWORK FOR THE FUEL CHANNEL

1 **How can one network learn the steam fraction, temperature and current density along the channel at every operating point?**

The network takes the position along the channel, the cell voltage and the inlet temperature, and returns the three fields. It is trained over 1.15 to 1.35 volts and 750 to 850 degrees until the three laws hold at points over the channel and the operating range; the inlet values are built into the form of the solution, and the current is kept below the limiting current. Any operating point is then read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

2 **Why does the local current make the channel harder to solve, and how is the network's answer checked?**

The current has no equation of its own along the channel: at every point it is the root of the charge transfer at the local steam fraction and temperature, so no field can be solved first. The network takes it as a third residual, in units of 50 millivolts so it weighs like the balances. Scored against a boundary-value solver at twelve operating points, it is within 0.0022 of the steam fraction and 0.19 kelvin, three times closer than finite volumes on 50 cells. *Ex_10.2 sections 3 – 5*

3 – 4 · THE OPTIMAL VOLTAGE

3 **How can the optimal cell voltage be found with a network, and what does the gradient save?**

The cost of an hour, the electricity bought less the hydrogen sold plus the share of the stack aged, is read from the network's fields, with the 10 kelvin limit as a penalty. Autograd through the network gives its derivative in the voltage, and gradient descent finds 1.250 volts in seconds; without the limit, 1.167 volts, where the cell cools 25 kelvin. A sweep of 81 solver runs agrees. The prices and the ageing law are assumptions, so the optimum is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

4 **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of voltage and temperature: what produces more now ages the cell faster. A solver gives the cost, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives every gradient by automatic differentiation. The degradation constants here are assumed, and published rates span an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

NEXT

L11.1 — Vision-Based Navigation

A network in a real-time control loop on a robot car, where the model is learned from camera frames and there is no PDE at all.

BEFORE THE NEXT SESSION

read	Liu ch. 5
run	Ex_10.2, sections 1 to 6
bring	your optimal voltage, and its worth
think about	what was physics, what was price

Block ten is finished: lithium in a particle, and steam in a ceramic cell - the course's last partial differential equations.